

CLARA POLLOCK

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EDUCATION

MPhys Astrophysics, University of Edinburgh, UK

September 2019 - May 2024 (expected)

- Grade Average 2021/2022: 75%
- Relevant courses: Fourier Analysis and Statistics (A2), Computer Modelling (A3), Observational Astronomy (A3)

Erasmus exchange, Lund University, Sweden

August 2022 – June 2023

- Grade Average 2022/2023: 74.5%
- Relevant Courses: Stellar Structure & Evolution (A), High Energy Astrophysics (A), Computational Astrophysics (A)

Clydeview Academy

2013 – 2019

- Advanced Higher: AAA in Chemistry, Mathematics, and Physics
- Higher: AAAAA in Chemistry, English, Mathematics, Physics, and Spanish

RESEARCH PROJECTS

MPhys Astrophysics Project

2023-2024 (8 months)

Exploring nebular diagnostics to infer metallicity of high-redshift galaxies

Supervisors: Dr Fergus Cullen & Dr Karla Arellano-Cordova

- Used strong line methods to determine temperature, density and metallicity using the direct method for CLASSY ($z=0$) survey, and early JWST data.
- Used Bayesian statistics (nested sampling) to ensure calculations of properties were robust.
- Constructing a new strong line metallicity calibration using these measurements that can be applied at high redshifts for galaxies with no auroral line measurements.

Summer Project, Institute for Astronomy, Edinburgh

2023 (6 weeks)

Spatially Resolving Properties of a Nearby Dwarf Galaxy

Supervisor: Macarena García Del Valle Espinosa

- Wrote Python script to fit emission lines for each spatial pixel of the galaxy.
- Evaluated new and robust estimates for metallicity in various star formation regions and their surrounding diffuse gas.
- Calculated velocity and velocity dispersion giving insight into rotation of galaxy, and origin of ionized gas.

Summer Project, Institute for Astronomy, Edinburgh

2022 (6 weeks)

Working with the first data from JWST

Supervisor: Dr Adam Carnall

- Created RGB colour images of high redshift CEERS galaxies observed in the first data release of JWST.
- Calculated properties such as galaxy size using Sérsic fitting.
- Published two papers as co-author, and credited for creating the [press image](#) of high-redshift galaxy candidate CEERS-93316.

PUBLICATIONS

- A.C Carnall et. al., “A first look at the SMACS0723 JWST ERO: spectroscopic redshifts, stellar masses, and star-formation histories”, [MNRAS 518, L45-L50](#) (2023)
- A.C. Carnall et. al., “A surprising abundance of massive quiescent galaxies at $3 < z < 5$ in the first data from JWST CEERS”, [MNRAS 520, 3974-3985](#) (2023)

SKILLS

- **Technical:** LaTeX, Python, Linux, Excel
- **Interpersonal:** Collaborative, approachable, effective communicator, responsible, independent learner.
- **Research:** Effective time management, skilled at problem solving, enthusiastic, reliable, organized.
- **Languages:** English (native), Swedish (A3)

WORK EXPERIENCE

Sales Assistant, Papyrus Gifts Glasgow

September 2018 – December 2021

- In my role at a small but busy gift shop I was responsible for sales, processing transactions, interacting with and helping customers, and sorting orders and deliveries. I developed skills in customer service, teamwork, and time management.

Coaching Assistant, Inverclyde Athletics Club

2016-2020

- For four years I volunteered as a coach at my local athletics club. I was regularly responsible for up to eight children, including for children of key workers during the pandemic. Coaching has allowed me to develop skills in leadership, communication, and confidence, as well as proving myself to be always patient, enthusiastic, and reliable.

SCHOLARSHIPS AND AWARDS

- DHL UK Foundation Undergraduate Scholarship (2019 - present)
- Certificate of Merit School of Physics and Astronomy (2019/2020, Year 1 and 2020/2021, Year 2)
- Dux Award Clydevew Academy (2017/2018, S5)
- Texas Instruments Excellence in Stem Award (Clydevew Academy 2016-2019)
- ABRSM Grade 7 Piano & Grade 6 Violin
- Duke of Edinburgh Bronze, Silver and Gold Award

HOBBIES

- Running, hillwalking, and hiking
- Painting and illustration
- Music (playing piano and violin)